Anthony Schurle

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Research Interests

Logic in Computer Science; Descriptive Complexity; Finite Model Theory

Education

Rice University Houston, Texas
B.S. Mathematics, B.A. Computer Science Aug. 2024 – May 2028
GPA: 3.8/4.00

Northland Christian School, Texas Aug. 2020 – May 2024
Valedictorian

Honors & Awards

Trustee Distinguished Academic Scholarship ($80,000), Rice University Apr. 2024
President's Honor Roll, Rice University Fall 2025

Relevant Coursework

Graduate & Cross-Listed: CS Logic, General Topology
Reading Course: Descriptive Complexity with Moshe Vardi
Undergraduate: Reasoning About Algorithms, Abstract Algebra I, Honors Analysis, Number Theory, Potpourri in Extremal Graph Theory, Honors Linear Algebra, Honors ODEs, Honors Statistics
Independent Study: *An Introduction to Mathematical Logic* (Enderton), *How to Prove It* (Velleman)

Professional Experience

Teaching Assistant | *Algorithmic Thinking (COMP182)* Jan. 2026 – May 2026
George R. Brown School of Engineering and Computing, Rice University

Teaching Assistant | *Computational Thinking (COMP140)* Aug. 2025 – Dec. 2025
George R. Brown School of Engineering and Computing, Rice University

Projects

Graft (Interactive Graph Theory Sandbox) | *JavaScript, SVG*

- Developed a Desmos-style web application for building and exploring graphs through an intuitive visual interface
- Implemented real-time computation of graph properties (degree sequences, connectivity, bipartiteness, chromatic number)
- Created export functionality for set notation, adjacency matrices, and LaTeX TikZ code

r2a (Disassembler) | *Python, RISC-V*

- Developed a formal specification for RISC-V instruction decoding using decision-tree optimization
- Implemented a testing pipeline to verify binary equivalence

Epidemiological Transmission Mapping | *Python*

- Modeled bacterial infection spread as a complete weighted directed graph using genetic (Hamming distance) and epidemiological data
- Implemented greedy algorithms to infer a rooted directed minimum spanning tree (RDMST) representing likely transmission pathways

Technical Skills

Languages: Python, C, Java, JavaScript, RISC-V Assembly, x86-64 Assembly
Tools: LaTeX, Git & GitHub, Docker, Linux